Maths Exploring for Terminal 10 Solutions

Corrected MCQs

1. **B.** $1 \times 2 + 2 \times 0 - 1 \times 3 = -1$. The scalar product is a real, not a vector.
2. **D.** The sum of the squares is 9; the norm is its square root, that is 3.
3. **A.** Two non-colonial directions generate the direction of the plan.
   directions not perpendicular to the vector.
4. **C.** The angle cosine is negative since the norms are positive.
   between $\pi/2$ and π.
5. **B.** Orthogonality is about their directions. In addition, perpendicular lines are stinging.
6. **D.** The point belongs to the plane and the vector that connects it to M is vertical, so normal to the plane.
7. **A.** The triangle MHP is right-angled at H where $P \neq H$; · the identity remains valid for $P = H$. It proves $MP \geq MH$.
8. **C.** The volume is one third of the product of the base area by height: 6 $\times 4/3 = 8$.
9. **B.** The scalar product is the angle cosine since the norms are 1; the angle is not oriented
   $2\pi/3$.
10. **D.** Write $H = A + t(1;1;0)$. Orthogonality $(M - H) \cdot (1;1;0) = 1 - 2t = 0$ given $t = 1/2$.

True/false. False: the projectee achieves the minimum, by Pythagoras. False: $(1;0;0)$ and $(0;1;0)$. True: each height must correspond to the chosen base.

Corrected applications and support

Application 1. Output 1, norms $\sqrt{2}$ and$\sqrt{2}$, therefore cosine $1/2$, angle $\pi/3$. The coordinated formula requires an orthonormal coordinate system.

Application 2. In the plan $z = 0, \vec{\imath}\cdot\vec{\jmath} = 0$, but $\vec{\jmath}$ belongs to the plan and is not normal. orthogonal directions are orthogonal; they are perpendicular if they also cut.

Implementation 4. $H = (t;t;0)$ and $\overrightarrow{MH}\cdot(1;1;0) = 2t-5 = 0$, therefore $H = (5/2;5/2;0)$. $MH^2 = 1/4 + 1/4 + 1 = 3/2$; distance $\sqrt{6}/2$.

Application 5. $H = (1;2;0)$; $\overrightarrow{HM} = (0;0;3)$ is orthogonal in both directions $\vec{\imath},\vec{\jmath}$. For $Q = (x;y;0)$, $MQ^2 = (x-1)^2 + (y-2)^2 + 9 \geq 9$, equality only in H.

Implementation 6. $AB = \sqrt{52}$, $AC = \sqrt{45}$, product 36. Cosine is worth 36 $/\sqrt{2340}$; the sine is worth $\overline{2340 - 1296}/\sqrt{2340} = 6\sqrt{29}/\sqrt{2340}$. Thus, it is 3 $\overline{29}$ and height 12/ $\sqrt{29}$.

Implementation 7. He's missing MH orthogonal plan (or $\overrightarrow{MH}$ colinear to a normal). $M = (1;2;3)$, H=(1;2;0) is suitable but Q=(0;0;0), yet in the plan, is not the projected.

Driving

Search the height in the figure before calculating a volume. For angle calculation, require verification cos $\theta \in [-1;1]$. Keep an exact expression before approximation.

Implementation 3. The midpoint is I=(1;1;0) and a normal vector is (2;2;0). The locus is the plane through I with this Normal. $MA^2 - MB^2 = 4x + 4y - 8$; therefore x+y=2. For example (1;1;0) and (2;0;3) agree. all points of the plan, without restriction on z.

Conduct and correction

Benchmarks for the teacher

Start with the three writings of the scalar product. The orthogonal lines are not necessarily pruning: " perpendicular" requires the intersection. To demonstrate that a right is orthogonal to a plane, choose two non-collinear directions. The general formula of distance point-plane is not required by heart: prefer the projected. Cartesian equations used in S54 progression can be announced by vector condition, then Formalised in S67.

Corrected activity

1. One item H of d written $H = A + t\vec{u}$, therefore $H(1+t, t, 0)$ for $t \in \mathbb{R}$.

2. $\overrightarrow{MH} = (1+t, t-2, -3)$. The scalar product with $\vec{u} = (1,1,0)$ value $1+t+t-2 = 2t-1$. Sound cancellation given $t = \frac{1}{2}$ and $H(\frac{3}{2}, \frac{1}{2}, 0)$.

3. The coordinate system shall be orthonormal: $MH^2 = (\frac{3}{2})^2 + (-\frac{3}{2})^2 + (-3)^2 = \frac{27}{2}$. So $MH = \frac{3\sqrt{6}}{2}$.

4. For $P \in d$, $\overrightarrow{HP}$ is colinear to $\vec{u}$ and thus orthogonal to $\overrightarrow{MH}$. The theorem of Pythagoras gives $MP^2 = MH^2 + HP^2 \geq MH^2$, with equality only if $P = H$. The calculated point is very unique point of the nearest right to M.

E1. In the orthonormal coordinate system, the scalar product is $2 \times 1 + (-1) \times 4 + 3 \times 0 = -2$.

E2. The scalar product is $3 \quad a + 2(-1) + 1 \times 4 = 3a + 2$. Orthogonality is equivalent to its cancellation: $3a + 2 = 0$, therefore $a = -\frac{2}{3}$.

E3. The two directors are non-collinear and the two scalar products are zero. $I(1,1,1)$. The location sought is the plan passing through I and normal to $\overrightarrow{AB} = (2,2,2)$, so normal $(1,1,1)$.

E4. The two vectors are non-zero, norms $\sqrt{2}$, and their scalar product is 1. So their angle $\theta \in [0, \pi]$ check cos $\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2}$either $\theta = \frac{\pi}{3}$.

E5. The area of the base rectangle is $\frac{3\times 4}{2} = 6$ cm^2. The height perpendicular to this base is 5 cm, so $V = \frac{6\times 5}{3} = 10$ cm^3.

Corrected from DM11

1. $\overrightarrow{AB} = (-2,2,0)$, $\overrightarrow{AC} = (-2,0,2)$ and $\overrightarrow{BC} = (0,-2,2)$; The three norms are worth $2\sqrt{2}$. The triangle is Its height is equal to $\sqrt{(2\sqrt{2})^2 - (\sqrt{2})^2} = \sqrt{6}$; its area is therefore $\frac{(2\sqrt{2})(\sqrt{6})}{2} = 2\sqrt{3}$.

2. The vectors $\overrightarrow{AB}, \overrightarrow{AC}$ their scalar products with $(1,1,1)$ value respectively $-2+2 = 0$ and $-2+2 = 0$. The non-zero vector $\vec{n}$ is therefore normal to (ABC).

3. The project O belongs to the right of director $\vec{n}$ passing through O, therefore $H = (t,t,t)$. Its condition of membership in the Eastern Plan $(t-2)+t+t = 0$, either $t = \frac{2}{3}$. We get $H(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$, which verifies the two conditions of a project.

4. $OH = \sqrt{3(\frac{2}{3})^2} = \frac{2}{\sqrt{3}}$. Taking the base ABC, volume is $V = \frac{(2\sqrt{3})(\frac{2}{\sqrt{3}})}{3} = \frac{4}{3}$ units of volume. Other base: the triangle OAB is right-angled at O, d ·aire $\frac{2\times 2}{2} = 2$; the right (OC) is perpendicular on the plan (OAB), and $OC = 2$. Thus $V = \frac{2\times 2}{3} = \frac{4}{3}$.

Corrected work in the manual or an anal

Exercise 23. In the given orthonormal coordinate system, the scalar product is the sum of the products corresponding coordinates:

$$(-3{,}5)(-4) + 2 \times 5{,}5 + (-4)(-3) = 14 + 11 + 12 = 37,$$

$$\frac{2}{3}\left(-\frac{3}{4}\right) + \frac{15}{7}\frac{14}{3} + \left(-\frac{12}{11}\right)\left(-\frac{22}{4}\right) = -\frac{1}{2} + 10 + 6 = \frac{31}{2}.$$

Exercise 27. In each case, the two vectors $\vec{u}$ and $\vec{v}$ given lead the plan and are not
A non-zero vector $\vec{w}$ is normal if, and only if, it is orthogonal to each of them.

1. $\vec{w} \cdot \vec{u} = -132 + 312 - 180 = 0$ and $\vec{w} \cdot \vec{v} = -66 + 546 - 480 = 0$. The vector $\vec{w}$ is normal in the plan.
2. $\vec{w} \cdot \vec{u} = 20 + 6 - 26 = 0$, but $\vec{w} \cdot \vec{v} = 18 - 6 - 10 = 2 \neq 0$. It is not normal in the plan.
3. $\vec{w} \cdot \vec{u} = \frac{1}{2} + 6 - 7 = -\frac{1}{2} \neq 0$. This single non-zero product is sufficient to exclude normality.